2.6 Proving Geometric Relationships

Geometry 2.6 (30-minute plan) · Student sheet

Name _____ Period ____ Date _____

I can…

- use the Right Angles Congruence Theorem.
- use the Congruent Supplements and Congruent Complements Theorems.
- prove and use the Vertical Angles Congruence Theorem.

Vocabulary · fill in the missing word

Flowchart proof — a proof that uses boxes and arrows to show the flow of a logical _____

Paragraph proof — a proof written as _____ in a paragraph

Part 1 · Notes and examples (fill in each note, then work the example)

Right Angles Congruence Theorem: all right angles are _____.

Example 1

∠ABC and ∠DEF are right angles. What can you conclude, and why?

Congruent Supplements Theorem: if two angles are supplementary to the same angle (or to congruent angles), then they are _____. The Congruent Complements Theorem works the same way.

Example 2

∠1 and ∠2 are supplementary, and ∠3 and ∠2 are supplementary. m∠1 = 47°. Find m∠3 and explain in a paragraph proof.

Vertical Angles Congruence Theorem is proved with the _____ Postulate and the Congruent Supplements Theorem.

Example 3

Two lines intersect. Two vertical angles measure (5x + 10)° and (7x − 18)°. Find x and all four angle measures.

Part 2 · Partner practice

Partner practice

Two vertical angles measure (3x + 20)° and (5x − 10)°. Find x, the angle measure, and the measure of an angle that forms a linear pair with it.